Space Software: F´ Lab

Winter Semester 2024

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Outline

Flight Software

F´ Framework

Hands-on tutorial

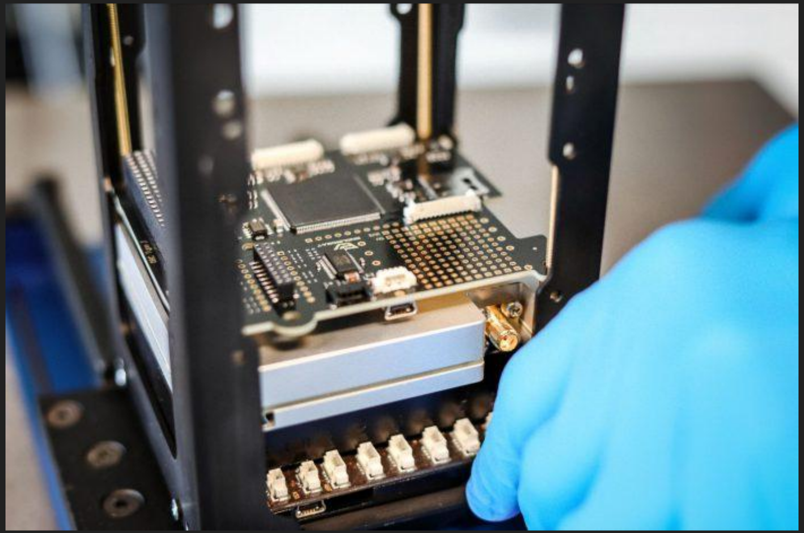
What is a Flight Software (FSW)

The FSW is what enables the spacecraft to perform all operations necessary to facilitate the science objective and perform maintenance tasks for the spacecraft.

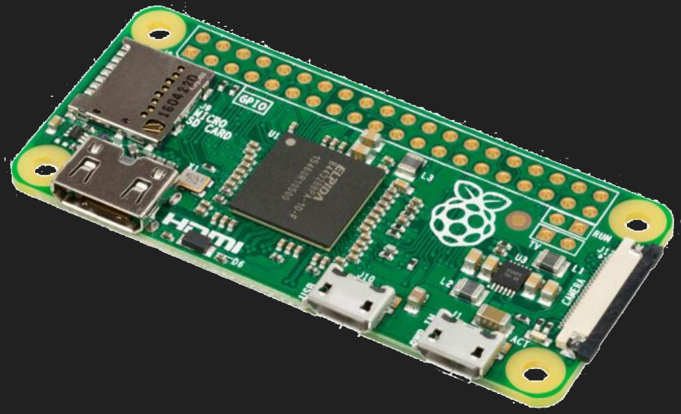
Important aspects:

- Safety
- Robustness
- Reliability
- Performance
- Modularity
- Maintainability

On-Board Computer (OBC)

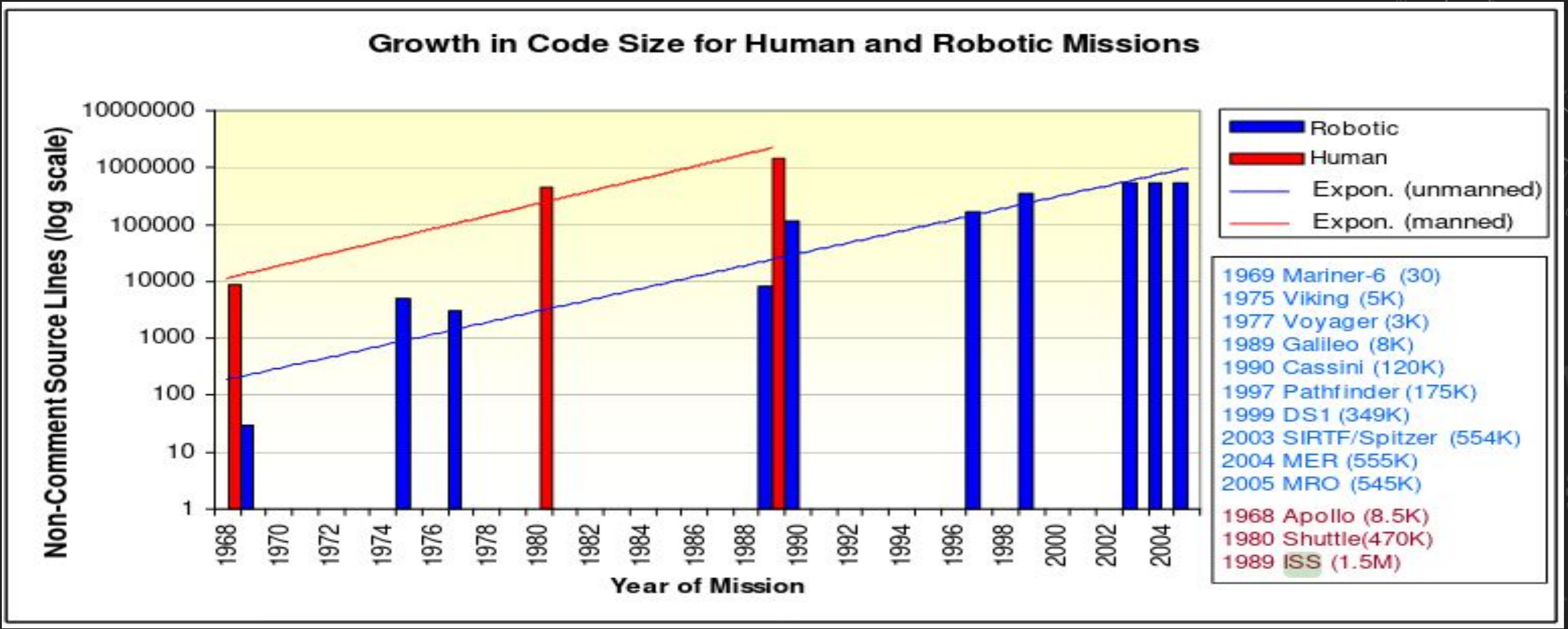


- A satellite contains from 1 or computers
- Controlling and communicating with all satellite subsystems, e.g.:
    - Payload
    - Electrical Power subsystem
    - Communication
    - ADCS



https://phxcubesat.asu.edu/subsystems/flight-software

Complexity of Flight Software



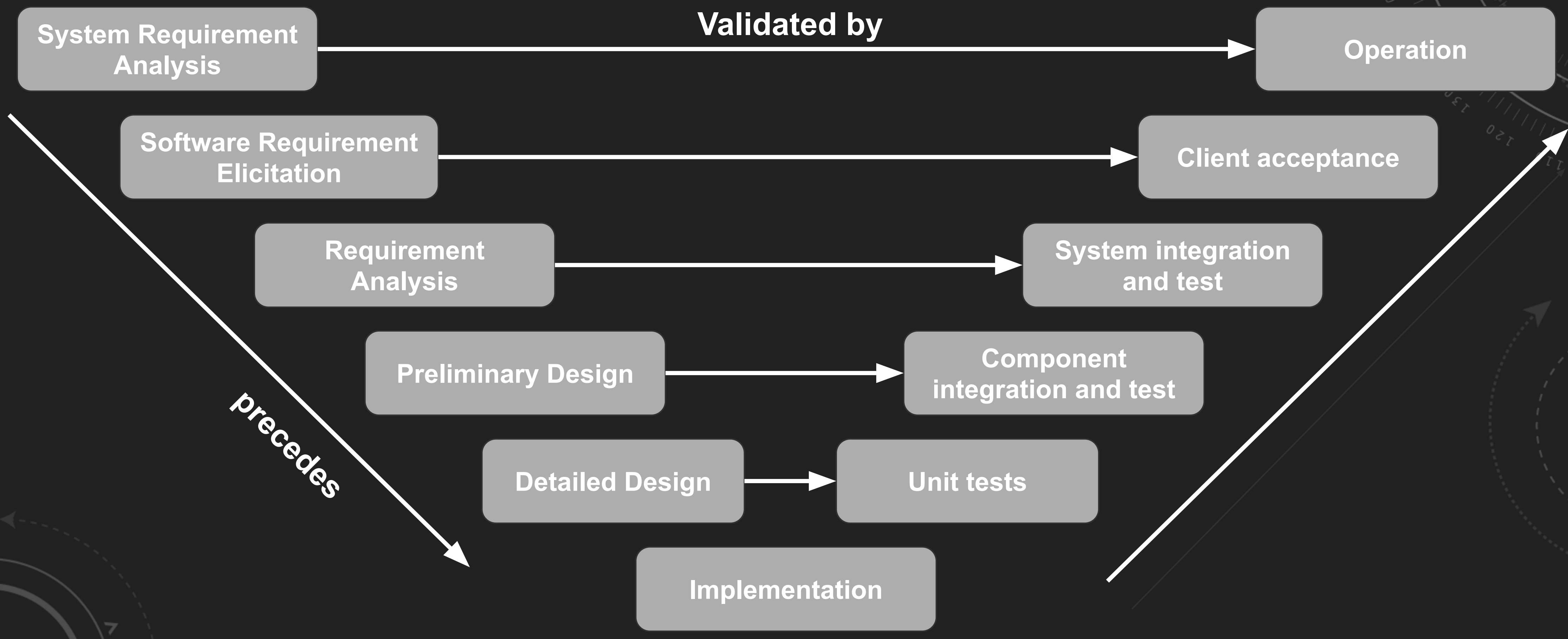
https://www.nasa.gov/wp-content/uploads/2015/04/418878main_fswc_final_report.pdf

Complexity of Flight Software

ISS Source Code lines:
Only for U.S. alone over 1.5m (FSW) on 44 computers communicating via 100 data networks transferring 400,000 signals

https://www.nasa.gov/international-space-station/space-station-facts-and-figures/

Development process - V model



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F´ Key Features

- **Component-Based Architecture**
  - Commands
  - Telemetry & Event generation
  - Parameters
- **Rapid Deployment**
  - Code Generation
  - Easy isolation of components (testing)
  - Ground Data System (testing)
- **Reusability**
  - Easy integration
  - Efficiency in development
  - Flight heritage
- **Portability**
  - Wide Compatibility
  - Operating System Flexibility
    - Linux, Baremetal, RTOS etc.

Project/Program	Type	OS	Hardware	Duration	Note
ISS-RapidScat	Instrument	-	-	Sep. 14 2014 - Nov. 18 2018	Target: Earth
ASTERIA	CubeSat/SmallSat	Linux	PowerPC	Aug. 14 2017 - Feb. 18 2020	Target: Exoplanets
MARS Helicopter (Ingenuity)	Airborne	Linux	Qualcomm Snapdragon 801	Jul 30 2020 - Ongoing	Target: Mars
Lunar Flashlight CubeSat	CubeSat/SmallSat	VxWorks	Sphinx (GR712 - LEON3FT SPARC)	Dec. 11 2022 - Ongoing	Target: Moon
Near Earth Asteroid Scout CubeSat	CubeSat/SmallSat	VxWorks	Sphinx (GR712 - LEON3FT SPARC)	Nov. 16 2022 - Ongoing	Target: Asteroids and Comets

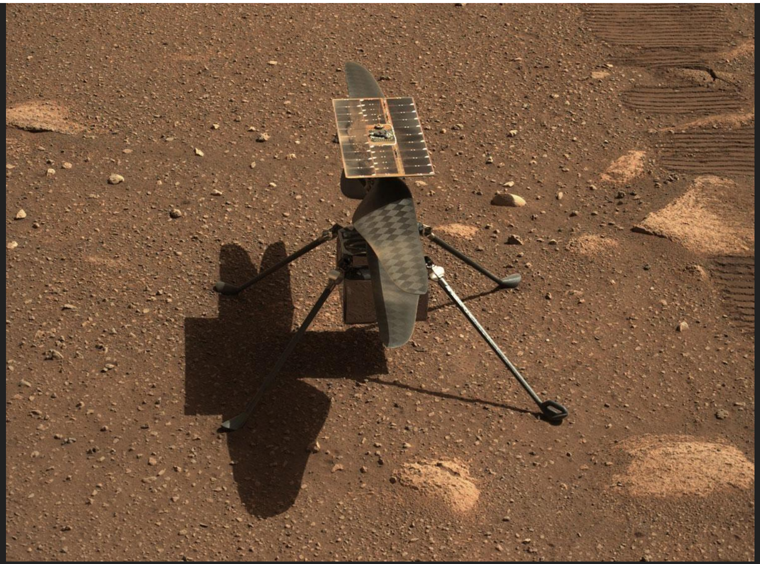
Future missions	
OWLS	Instrument
CADRE	Robot
COLDArm	Robotic arm

MARS Helicopter, Ingenuity

Flight Log

By the Numbers

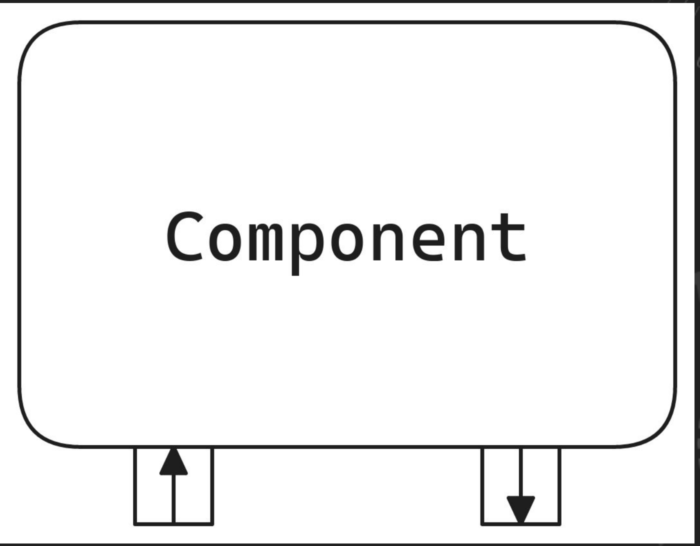
FLIGHTS	FLIGHT TIME	DISTANCE FLOWN	MAX. GROUND SPEED	HIGHEST ALTITUDE
72	~128.8 MINS	11 MILES	22.4 MPH	24 METERS
		(~17.0 km)	10 m/s	(~79 ft)



https://science.nasa.gov/mission/mars-2020-perseverance/ingenuity-mars-helicopter/

Component-based architecture

- Individual blocks
- Define the behavior of the system
- Well-defined interfaces
- Communication through ports



F´ Terminology

Component:

- Independent
- Not intercepting other components

Port:

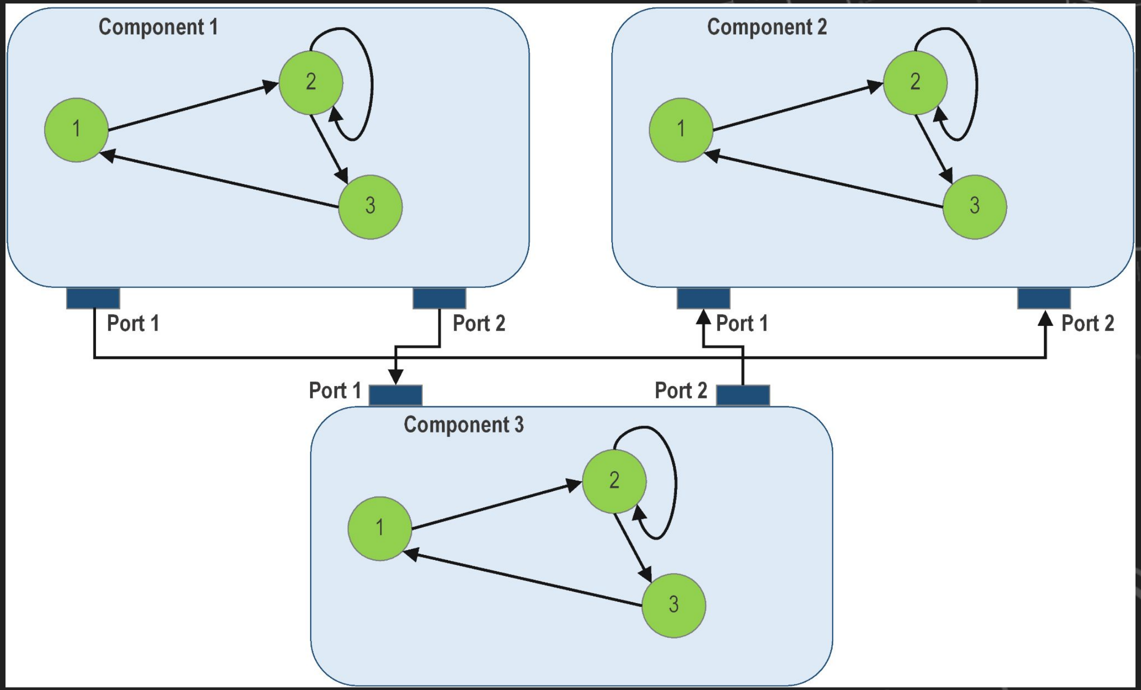
- represents a connection between Components
- acting as communication channels in F’
- interaction between components

Topology:

- Represents the full F´ system and all communication channels

Deployment:

- Represents the final product, the executable



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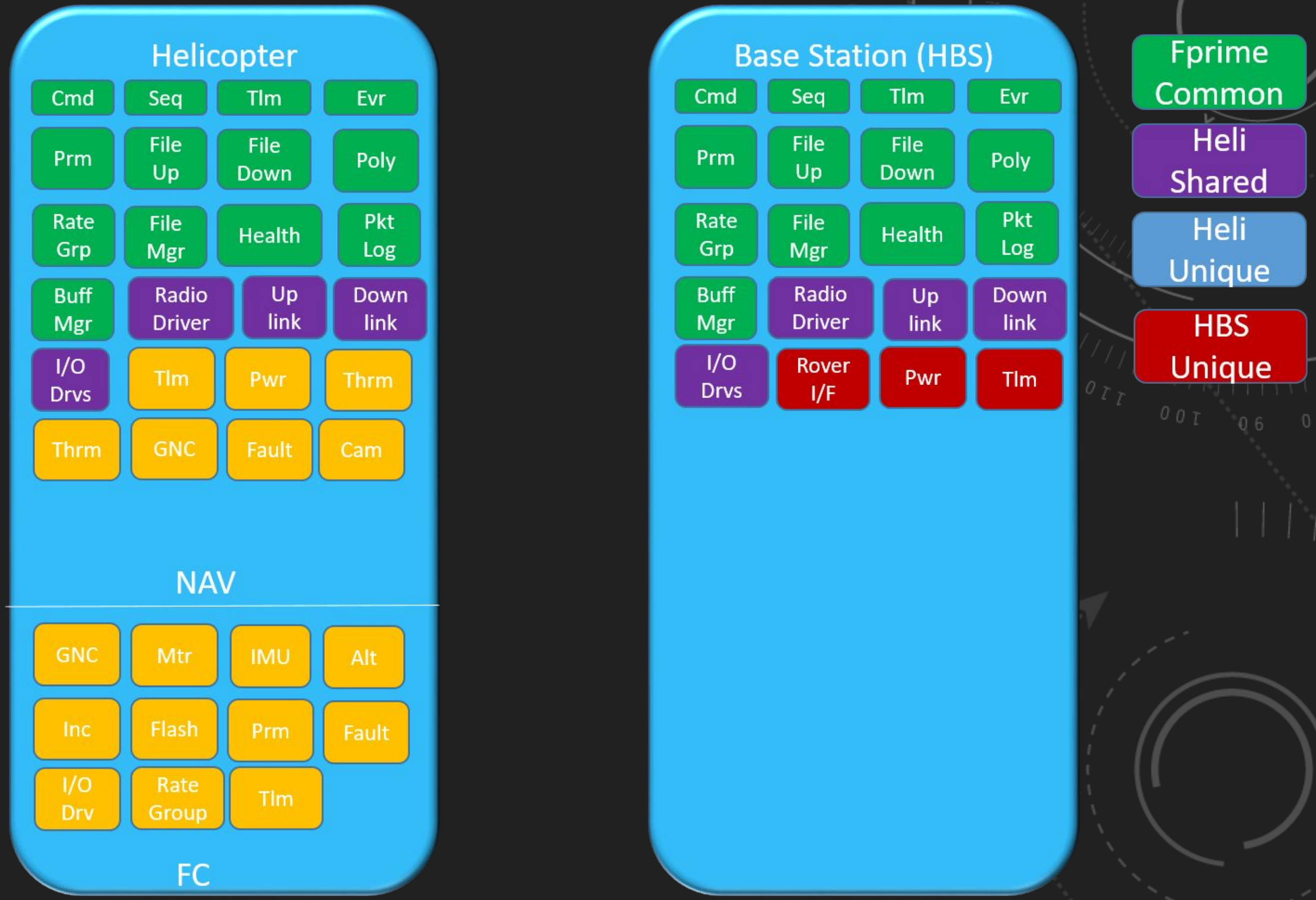


Mars Helicopter flight software components.

F´ Build System

Build:

- compile of code using CMake
- stores all by-products, generated code, and build products from an F´ build

```
$ tree
.
├── CMakeLists.txt
├── subbinary
│   ├── CMakeLists.txt
│   └── main.cpp
├── sublibrary1
│   ├── CMakeLists.txt
│   ├── include
│   │   └── sublib1
│   │       └── sublib1.h
│   └── src
│       └── sublib1.cpp
└── sublibrary2
    ├── CMakeLists.txt
    └── include
        └── sublib2
            └── sublib2.h
```

F´ Build System

Build:

- compile of code using CMake
- stores all by-products, generated code and build products from an F´ build

Toolchain:

- Tools to build in a various architectures
- Cross-Compilation

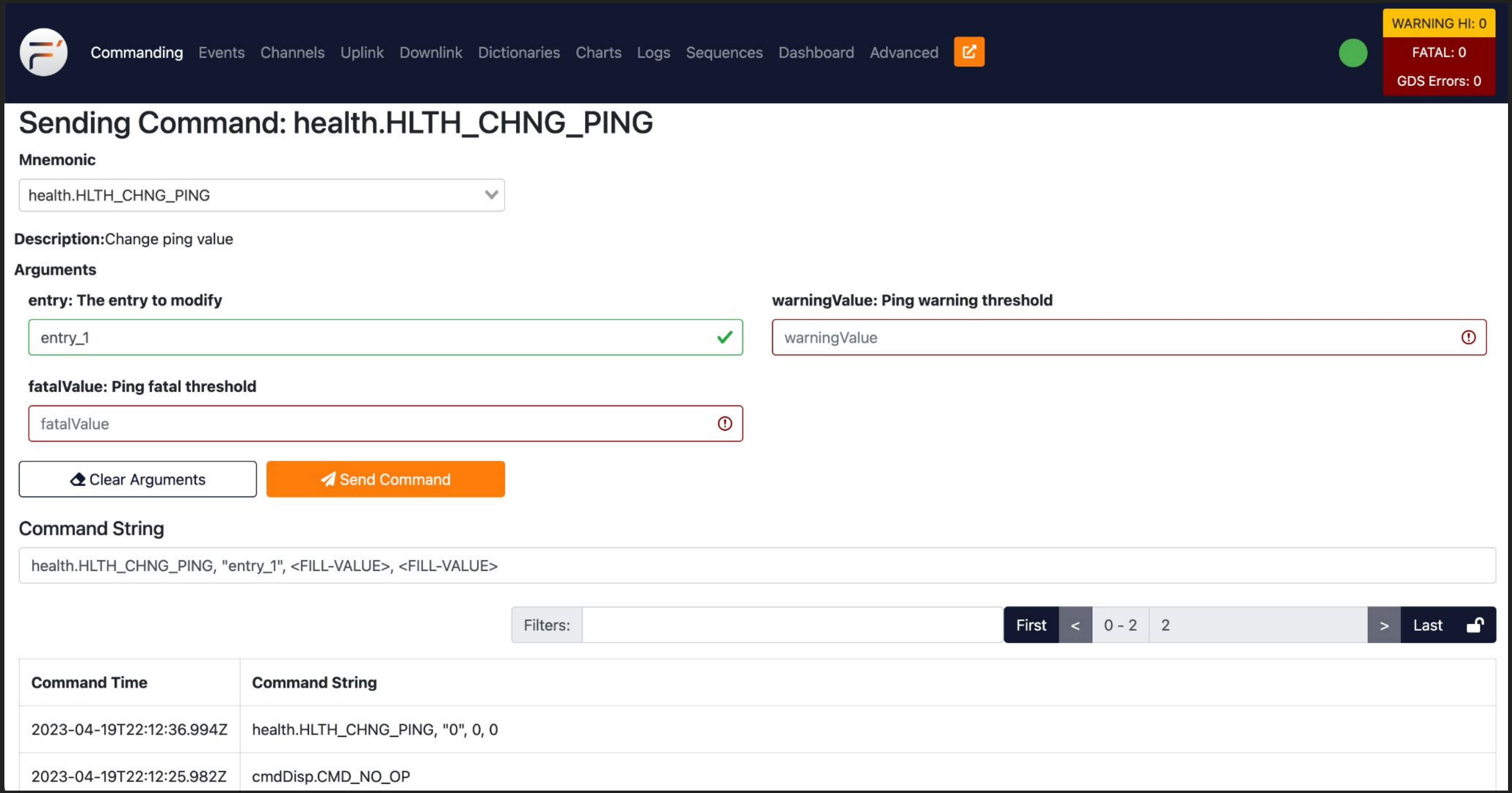
Board Name	Chip	Toolchain	Date Tested	F´ Version	arduino-cli Board Version	Upload Guide
Teensy 4.1	ARM Cortex-M7	teensy41	10/30/2024	3.5.0	teensy:avr@1.59.0	README
Teensy 4.0	ARM Cortex-M7	teensy40	10/30/2024	3.5.0	teensy:avr@1.59.0	README
Teensy 3.2	ARM Cortex-M4	teensy32	10/30/2024	3.5.0	teensy:avr@1.59.0	README
Raspberry Pi Pico	RP2040	rpipico	10/30/2024	3.5.0	rp2040:rp2040@4.2.0	README
Raspberry Pi Pico W	RP2040	rpipicow	10/30/2024	3.5.0	rp2040:rp2040@4.2.0	README
Raspberry Pi Pico 2	RP2350	rpipico2	10/30/2024	3.5.0	rp2040:rp2040@4.2.0	README
Adafruit KB2040	RP2040	kb2040	10/30/2024	3.5.0	rp2040:rp2040@4.2.0	README
Adafruit Feather RP2040	RP2040	featherrp2040	10/30/2024	3.5.0	rp2040:rp2040@4.2.0	README
Adafruit Feather RP2040 RFM	RP2040	featherrp2040rfm	10/30/2024	3.5.0	rp2040:rp2040@4.2.0	README
Adafruit Feather RP2350 HSTX	RP2350	featherRP2350	10/30/2024	3.5.0	rp2040:rp2040@4.2.0	README
Adafruit Feather M0	SAMD21 ARM Cortex-M0	featherM0	10/30/2024	3.5.0	adafruit:samd@1.7.16	README
Adafruit Feather M4	SAMD51 ARM Cortex-M4	featherM4	10/30/2024	3.5.0	adafruit:samd@1.7.16	README
Adafruit Feather ESP32 V2	ESP32	featherESP32	10/30/2024	3.5.0	esp32:esp32@2.0.9	README
Adafruit Circuit Playground Express	SAMD21 ARM Cortex-M0	circuitplayexpress	10/30/2024	3.5.0	adafruit:samd@1.7.16	README
Adafruit Circuit Playground Bluefruit	nRF52840	circuitplaybluefruit	10/30/2024	3.5.0	adafruit:nrf52@1.6.1	README
Adafruit Feather nRF52840	nRF52840	featherNRF52840	10/30/2024	3.5.0	adafruit:nrf52@1.6.1	README
Adafruit Metro M4 Grand Central	SAMD51 ARM Cortex-M4	grandcentral	10/30/2024	3.5.0	adafruit:samd@1.7.16	README
ESP32 Dev Module	ESP32	esp32	10/31/2024	3.5.0	esp32:esp32@2.0.9	README

Ground Data System (GDS)

Interaction with the embedded system

An easy integration testing environment

Ground Data System (GDS)



Ground Data System (GDS)

Commanding Events Channels Uplink Downlink Dictionaries Charts Logs Sequences Dashboard Advanced

WARNING HI: 0
FATAL: 0
GDS Errors: 0

Events

Filters: First < 0 - 6 6 > Last

Event Time	Event Id	Event Name	Event Severity	Event Description
2023-04-19T21:12:25.983Z	0x501	cmdDisp.OpCodeDispatched	COMMAND	cmdDisp.CMD_NO_OP dispatched to port 0
2023-04-19T21:12:25.983Z	0x507	cmdDisp.NoOpReceived	ACTIVITY_HI	Received a NO-OP command
2023-04-19T21:12:25.983Z	0x502	cmdDisp.OpCodeCompleted	COMMAND	cmdDisp.CMD_NO_OP completed
2023-04-19T21:12:36.995Z	0x501	cmdDisp.OpCodeDispatched	COMMAND	health.HLTH_CHNG_PING dispatched to port 5
2023-04-19T21:12:40.512Z	0x2005	health.HLTH_CHECK_LOOKUP_ERROR	WARNING_LO	Couldn't find entry 0
2023-04-19T21:12:40.512Z	0x503	cmdDisp.OpCodeError	COMMAND	health.HLTH_CHNG_PING completed with error VALIDATION_ERROR

Outline

Satellites & Flight Software

F´ Framework

Hands-on tutorial

Hands-on tutorial agenda

Setup

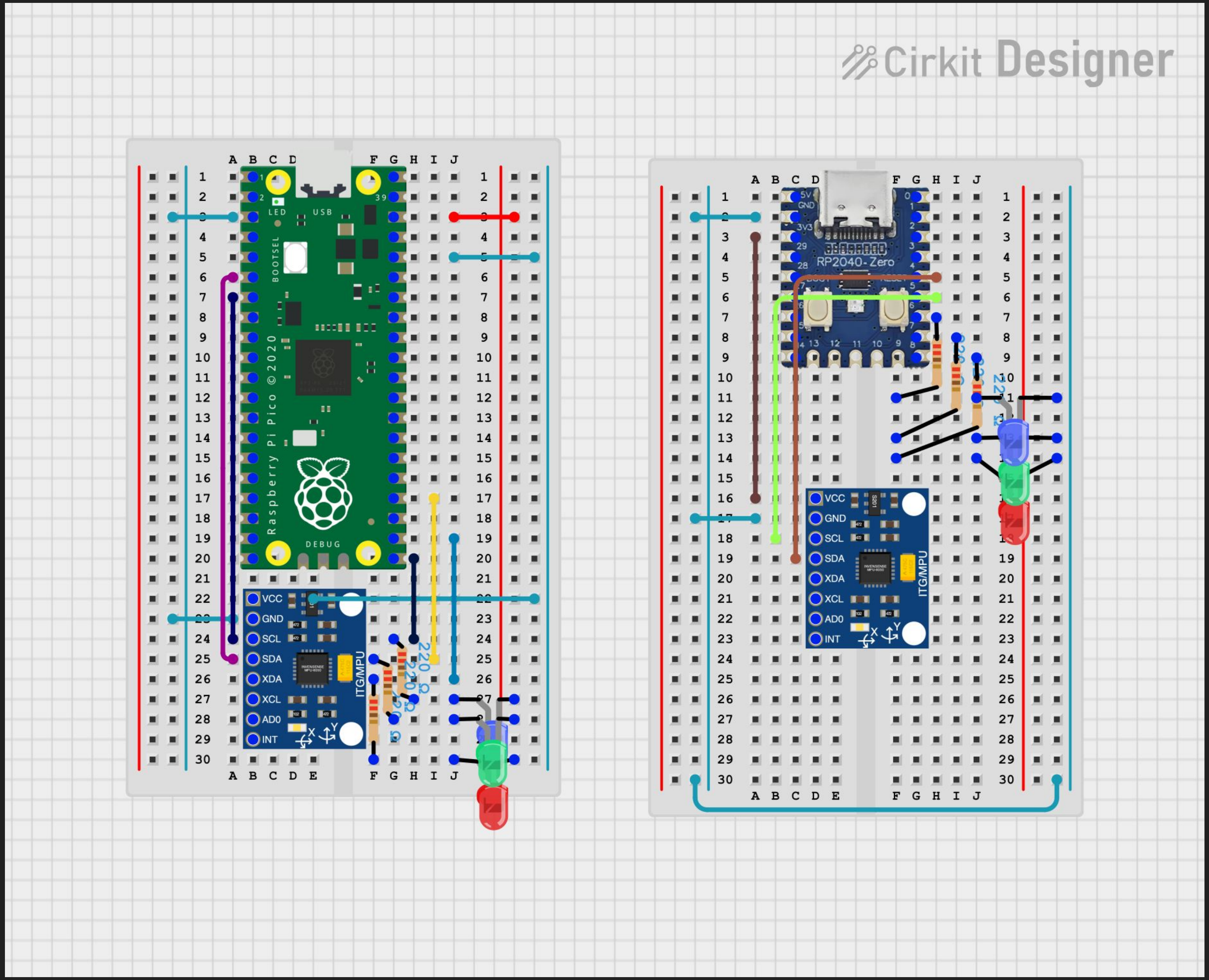
- ➢ Hardware
- ➢ Software

IMU Component Tutorial

- ➢ Introduction
- ➢ Defining ports and types
- ➢ MotionSensor Ports
- ➢ Arduino Example
- ➢ Motion Sensor Implementation
- ➢ Defining telemetry and events
- ➢ Developing logic in component
- ➢ Full System Integration

Hardware setup

- Raspberry Pi Pico
- MPU6050 via I2C
  - Serial Data Line (SDA)
  - Serial Clock Line (SCL)
- Red, Green and Blue LED
  - GPIO Pins
- 3x 220 Ohm resistors for the LEDs
- [Pico Series]()
- [RP2040-Zero]()



Software setup

- **Setup in lab or before**
    - Follow carefully the git repository steps
- **Linux Dependency**:
  F´ requires a Linux-based system for installation
- **Windows Users**:
  For Windows 10 or above users, Windows Subsystem for Linux (WSL) is recommended.
- **Alternative Virtualization Options**:
  Install Ubuntu on a virtual machine using software like VirtualBox, VMWare etc.

Hands-on tutorial agenda

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- ➢ Software

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